



School of Computing
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PRODUCT VISION DOCUMENT

KaliMind

Autonomous Vulnerability Assessment Agent

LLM-Assisted Vulnerability Assessment via Model Context Protocol (MCP)

Team KaliMind

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1. Audience

Who are the users of the product?

- **Primary Audience:** Security professionals, ethical hackers, and penetration testers who need to automate and accelerate vulnerability assessment workflows. Also includes CTF (Capture The Flag) participants and cybersecurity students learning offensive security techniques in authorised lab environments.
- **Secondary Audience:** Enterprise security teams and SMEs requiring scalable, repeatable security audits without a full in-house red team. Academic researchers exploring AI-assisted security tooling, and faculty evaluators assessing innovation and feasibility of this project at SRMIST.

2. Needs

What problems does the product solve for the audience?

- **Primary Needs:** Traditional penetration testing is manual, time-consuming, and requires scarce expert knowledge to chain tools, validate findings, and filter false positives. Existing scanners use rigid, signature-based detection and cannot reason about context or autonomously adapt their approach. Organisations face a global shortage of skilled penetration testers, creating a significant security coverage gap.
- **Secondary Needs:** Penetration testing tools such as nmap, Metasploit, Hydra, SQLMap, WPScan, and John the Ripper currently operate in silos, requiring manual data transfer between each stage. Security teams also need automated, structured vulnerability reports to communicate risk to management and demonstrate compliance.

3. Products

What is the product, and what does it offer?

- **Core Product:** KaliMind is an LLM-assisted, autonomous vulnerability assessment agent that bridges the reasoning power of Large Language Models (Claude via the Anthropic API) with the execution capabilities of industry-standard Kali Linux penetration testing tools. Using the Model Context Protocol (MCP), the LLM autonomously invokes, chains, and interprets results from tools including nmap, gobuster, dirb, nikto, WPScan, SQLMap, Metasploit, Hydra, John the Ripper, and enum4linux — all orchestrated without human intervention between steps.
- **Additional Features:** Context-aware tool chaining where the LLM adapts its strategy based on intermediate scan results. Raw shell command execution on the Kali machine. A health-check endpoint that verifies tool availability before a session begins. A modular Flask REST API allowing new tools to be registered without refactoring. Deployable on Kali VM, Docker container, or bare-metal. Automated structured vulnerability report generation at session end.

4. Values

What principles guide the product, and what makes it unique?

- **Core Values:** Authorised Use Only — KaliMind is designed exclusively for systems the operator owns or has explicit written authorisation to test. Security First — the Kali backend API binds to the local network interface by default and must never be exposed to the public internet. Intelligence over Brute Force — the LLM’s reasoning layer adapts attack strategy dynamically, reducing false positives and enabling context-aware exploitation.
- **Differentiators:** Unlike conventional static scanners, KaliMind enables the LLM to function as an active, reasoning penetration tester — autonomously chaining tools, interpreting intermediate results, adapting payloads, and generating actionable reports. The dual-timeout architecture ensures partial results are preserved even when tools time out, maximising intelligence gathered per session. The MCP architecture is modular and extensible with zero core refactoring required to add new tools.

5. Vision Statement

What is the overarching goal of the product?

“To democratize cybersecurity by empowering organisations of all sizes with an intelligent, autonomous vulnerability assessment agent that thinks like an expert penetration tester — making secure infrastructure accessible, scalable, and actionable for a safer digital world.”